

Fingerprint terahertz spectroscopy combined with machine learning for multicomponent mixture analysis

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ABSTRACT

The distinctive vibrational features in terahertz (THz) spectroscopy characterize a “fingerprint” of the single-component molecular substance. However, due to componential spectral overlapping and baseline drift, the identification and quantification of multicomponent mixtures are quite challenging for THz spectral analysis. A systematic and feasible strategy has been proposed by combining machine learning with THz spectroscopy for both qualitative and quantitative analysis. After the component number was effectively determined by singular value decomposition (SVD), nonnegative matrix factorization (NMF) and self-modeling mixture analysis (SMMA) were applied to extract componential THz spectra. The difficulties of NMF and SMMA encountered in handling ternary mixtures were solved. The results show component spectra extracted by SMMA are highly consistent with the experimental spectra of pure substances after standardization to correct baseline drift, which greatly facilitates rapid identification of compositions in mixtures. Additionally, compared to back-propagation neural network (BPNN), support vector regression (SVR) predict the contents of each individual component with high robustness and the decision coefficient R^2 greater than 0.949. Fingerprint terahertz spectroscopy enhanced by machine learning provided an effective strategy for mixture analysis in practical applications.

1. Introduction

With the dramatic progress of terahertz (THz) technology and years of extensive research, it has been verified that signatures in THz spectroscopy characterize a “fingerprint” of the single-component molecular substance [1–3]. These distinctive vibrational features of substance in the THz region are associated with intermolecular collective vibrational modes of the hydrogen-bonding network and low-frequency intramolecular vibrational modes of molecules [4,5]. Therefore, these features are highly sensitive to the structure and conformation of molecules, as well as the nearby environment. Various substances can be recognized directly and rapidly according to their distinctive THz features. Reagents of food additive shown in Table 1 were purchased from Aladdin Biochemical Technology Co. Ltd. The procedure of the sample preparation and THz measurement were consistent with the procedure described in the following Experiment Section. As shown in Table 1, the pure reagents of food additive measured in the experiment revealed

unique THz spectral features, which are completely different from each other, implying that they can be well distinguished from their characteristic THz spectra. It should be noted that each food additive revealed rich THz features, including significant and weak features. However, when all the spectra are plotted in one figure, the weak features became unnoticeable.

Due to complicated components, in addition to broad absorption features in the THz region, multicomponent mixtures exhibit componential spectral overlapping and baseline drift in their THz spectra [6–8]. It is difficult to obtain compositions and contents in mixtures directly, but effective solutions are often realized by means of machine learning methods [9–12]. For example, principal component analysis (PCA) was applied to separate samples into a ternary diagram for classification, and quantitative analysis was achieved by partial least-squares (PLS) regression and artificial neural networks (ANN) [10]. Hard modeling factor analysis (HMFA) was applied to resolve THz spectral matrices from mixtures [11]. Support vector regression (SVR)

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Table 1
THz spectra of food additives measured in the experiment.

THz spectra	Food additive	Formula	Category
	Sorbic acid	$C_6H_8O_2$	Antiseptic
	Potassium sorbate	$C_6H_7KO_2$	
	Benzoic acid	$C_7H_6O_2$	
	Sodium benzoate	$C_7H_5NaO_2$	
	VC	$C_6H_8O_6$	
	Citric acid	$C_6H_8O_7$	Antioxidant
	BHA	$C_{11}H_{16}O_2$	
	Propyl gallate	$C_{10}H_{12}O_5$	
	Aspartame	$C_{14}H_{18}N_2O_5$	Sugar substitute
	Xylitol	$C_5H_{12}O_5$	
	D-Glucose	$C_6H_{12}O_6$	Edulcorant
	D-Fructose	$C_6H_{12}O_6$	
	D-Sucrose	$C_{12}H_{22}O_{11}$	
	D-Lactose	$C_{12}H_{22}O_{11}$	
	Taurine	$C_2H_7NO_3S$	
	L-Lysine	$C_6H_{14}N_2O_2$	
	L-Tyrosine	$C_9H_{11}NO_3$	
	L-Histidine	$C_6H_9N_3O_2$	Nutrient supplement

and PLS were used for quantitative modeling on THz spectra of ternary isomer mixtures of saccharide [12]. However, problems still exist because these methods are based on the premise that the number of components k is assumed to be known before they are utilized, and also, just the qualitative or quantitative analysis for the multicomponent mixtures can be solved.

As shown in Fig. 1, a systematic and feasible strategy for both qualitative and quantitative analysis of multicomponent mixtures has been proposed by combining THz time-domain spectroscopy (THz-TDS) with machine learning in our previous work [13]. To develop the strategy for more complex ternary mixtures analysis, three commonly used food additives, preservative potassium sorbate [14], antioxidative butylated hydroxyanisole (BHA) [15] and edulcorator glucose, were mixed with different content ratios, and treated as “unknown” mixtures. The measured THz spectral data of the mixtures are listed by columns as spectral matrix $V_{m \times n}$ (m represents the number of data points for each spectrum, and n is the number of mixed samples). After the component number k was effectively determined by singular value decomposition (SVD) [13,16], nonnegative matrix factorization (NMF) [17,18] and

self-modeling mixture analysis (SMMA) [19,20] were applied to extract componential THz spectra from $V_{m \times n}$. However, for ternary mixtures, NMF was inadequate to resolve componential THz spectra even after pure-variable initialization. Compared to NMF, the spectra extracted by SMMA were more consistent with THz signatures of pure substances. But, the obvious baseline drift becomes another serious issue. To solve baseline drift, the measured spectral vector and the extracted spectral vector were standardized. Ultimately, the extracted component spectra are highly consistent with the THz spectra of corresponding substances. To quantify ternary mixtures, SVR [21–23] and more nonlinear back-propagation neural network (BPNN) [24,25] were applied to predict the component content of the complex mixtures. As a result, SVR has been demonstrated to achieve high accuracy in quantitative analysis [26], especially for the small training dataset. With improvements in the strategy for identifying and quantifying ternary mixtures, a thorough in-depth approach for multicomponent mixture analysis was provided in this work, and THz spectroscopy enhanced by machine learning methods can be a reliable and effective strategy for mixture analysis in practical applications.

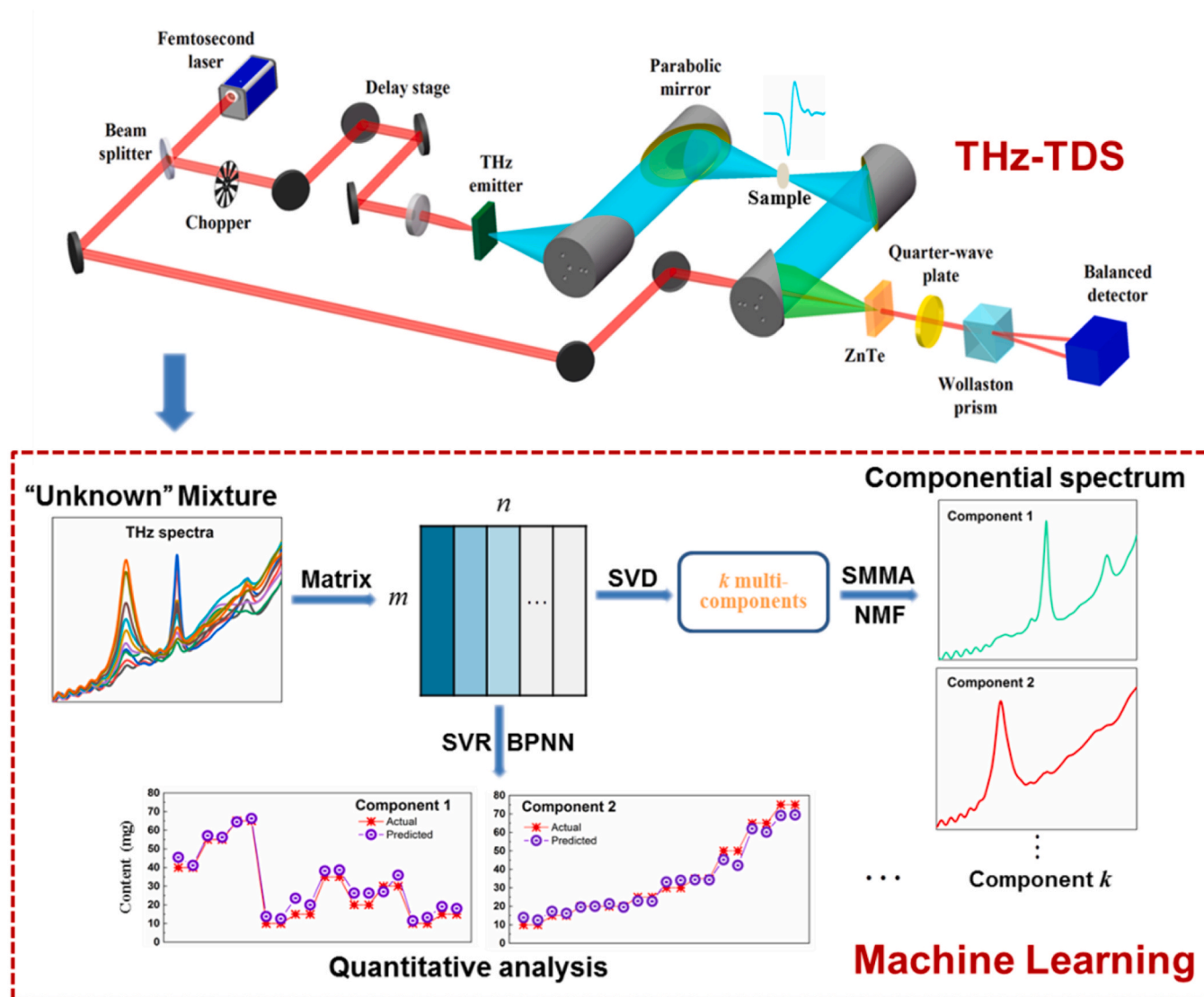


Fig. 1. Schematic diagram of the experimental system and machine learning processing.

2. Experiment

2.1. Reagents and sample preparation

Potassium sorbate (CAS: 24634–61–5, purity $\geq 99.0\%$) and butylated hydroxyanisole (CAS: 25013–16–5, purity $\geq 99.0\%$) were purchased from Aladdin Biochemical Technology Co. Ltd. Glucose (CAS: 50–99–7, purity $\geq 99.0\%$) was purchased from Yobios Biotechnology Co. Ltd. The granular reagents were finely ground. Their molecular structures and THz spectra are displayed in Fig. 2(a). It is clearly shown that the THz spectral characteristics can be well distinguished for each reagent. Table 2 shows a comparison of the peaks/bands of PS, BHA and Glu measured in this work with those described in the literature. The significant absorption peak of PS at 0.97 THz is in well agreement with the reported literature [27,28], and this peak is usually utilized as the major feature to distinguish PS in agricultural products and milk powder [27, 28]. The weak absorption bands of PS at 1.68 and 1.99 THz have not been reported in the literature due to the interested measurement range below 1.6 THz [27,28]. Moreover, the absorption features of BHA and Glu are highly consistent with those reported in the literature [29–31].

For sample preparation, as shown in Fig. 2(b), 10 ternary samples (S1–S10) with different contents of three pure reagents were successively marked in the triangle. Here, taking S5 as an example, it contained 25%

PS, 60% BHA and 15% Glu in a total mass of 100 mg. And also, 100 mg polytetrafluoroethene (PTFE) was added as the diluent and adhesive, which is of high transparency in the THz region. Then, all the weighted reagents were fully mixed to ensure spectral homogeneity. Finally, they were compressed into a tablet with 13 mm diameter at hydraulic pressure of 7 MPa. To acquire a sufficient dataset and ensure reproducibility in measurement, three replicated tablets for each mixed percentage were prepared for SVR training and testing. Additionally, samples with binary mixtures (containing any two of PS, BHA and Glu) were also prepared using the same procedure as described in our previous work [13].

2.2. Terahertz measurement

The samples were measured by a Teraview THz spectrometer in a purified dry-air atmosphere. They were scanned 1800 times, and all the signals were averaged to lower the noise. The time-domain waveform was Fourier transformed to the frequency-domain waveform. Then, the absorption data were obtained from the logarithm ratio between the sample and reference in the frequency-domain spectra, which were finally used for further analysis. Due to the strong absorption of molecular solids (especially in high content) and gradually decreased signal-to-noise ratio (SNR) (especially in high frequency), THz

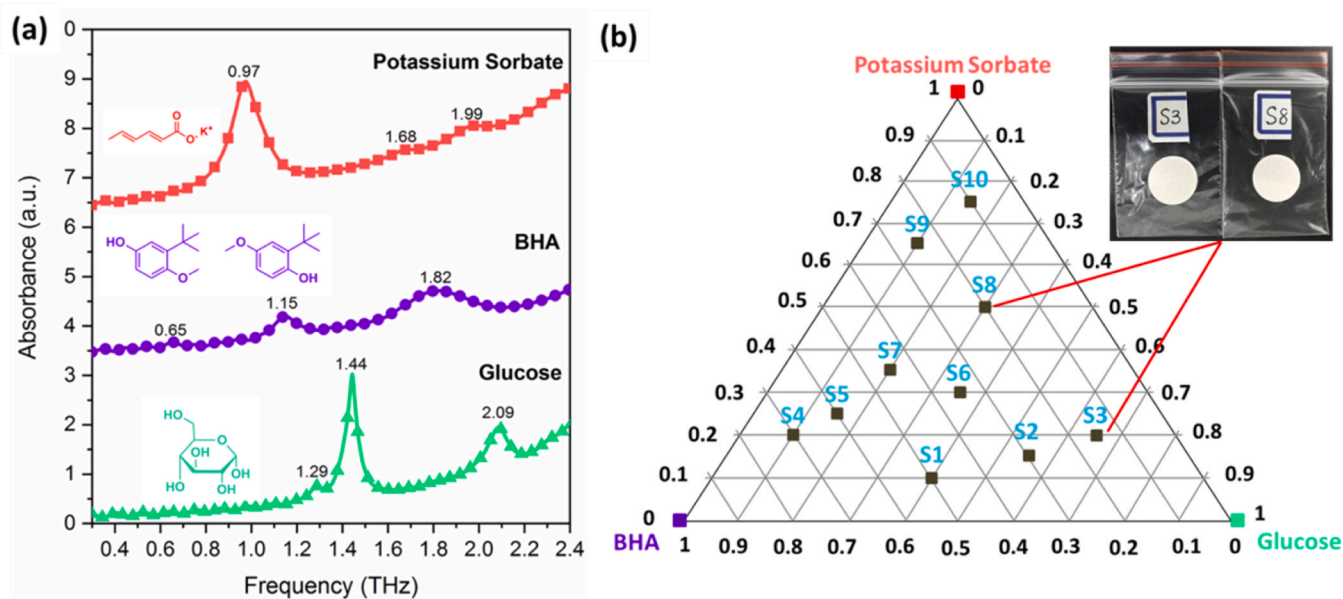


Fig. 2. (a) THz spectra of potassium sorbate (PS), BHA and glucose (Glu) with their molecular structures. (b) Ternary diagram displaying 10 mixed samples. The three vertices correspond to reagents of PS, BHA, and Glu, and each sample is mixed with these three reagents. The value 1 represents 100% content (100 mg), values among 0 and 1 represent the relative percentage in content of the related reagents. The inserted image shows the compressed tablet.

Table 2

The main peaks/bands of PS, BHA and Glu measured in this work compared to those described in the literature.

Main peaks/bands (THz)			
PS			
This work	0.97		1.68
Ref.[27]	0.97		1.99
Ref.[28]	0.97		
BHA			
This work	0.65	1.15	1.82
Ref.[29]	0.64	1.14	1.81
Glu			
This work	1.29	1.44	2.09
Ref.[30]	1.26	1.45	2.10
Ref.[31]	1.29	1.44	2.10

signatures of the samples above 3 THz become blurry and indistinguishable. Therefore, each spectrum was confined to the range of 0.3–2.4 THz with a spectral resolution of 3 GHz, and $m=701$ data points were acquired.

To acquire the SVR training dataset, two identical tablets prepared as described above were measured twice at two different positions along both surfaces in consideration of the influence of the scattering path. Therefore, eight spectra were obtained for each tablet. Considering the mixed percentage $n=10$ in this work, eighty sets of spectral data were used to train the SVR model, which established the relationship between the mixed spectral features and target content. Furthermore, the remaining set of tablets was measured once along both surfaces. Thus, twenty sets of spectral data were obtained and labeled from 1 to 20, which were further used as testing datasets to verify the SVR model.

3. Machine learning

3.1. Initialization of NMF

NMF is an efficient algorithm for approximating a high-dimensional matrix from two low-dimensional matrices [32]. It can mathematically decompose a nonnegative matrix $V_{m \times n}$ into two nonnegative matrices by a multiplicative iterative update algorithm up to approximation error [17,18,32]. Additionally, in the standard NMF algorithm, it commonly

begins with random initialization of the concentration matrix $H_{k \times n}$. For binary mixtures, several iterations from random initialization may achieve optimum decomposition. However, for ternary mixtures or more complicated mixtures, random initialization without limitations is subject to vast iterations, and unique spectral decomposition with physical interpretation is difficult to be obtained. To solve random initialization, the pure variable approach [19,20] of primary treatment in SMMA was applied to initialize NMF. Because the intensity of the pure variable is in direct proportion to the content of the corresponding constituent [33], it makes the initialization more straightforward and reasonably interpretable.

A pure variable (e.g. wavenumber) is contributed from only one constituent and can be statistically calculated by [20],

$$p_{ij} = \frac{\omega_{ij} \sigma_i}{\mu_i + \alpha} \quad (1)$$

where ω_{ij} corrects for previous pure variables and represents the weight function, α offsets the low pure variable for noise suppression and avoids dividing by zero. σ_i and μ_i are the standard deviation and the mean value of the variable i , respectively.

Therefore, after the component number k was pre-determined by SVD, the first k pure variables were selected from the spectral matrix $V_{m \times n}$. Then, the initialization of concentration matrix $H_{k \times n}$ was set to the normalized column values of $V_{m \times n}$.

Table 3
SVD result of matrix $V_{m \times n}$.

PS-BHA-Glu ^a Mixtures					
σ	82.2336	8.9178	5.1690	1.3450	0.5749
γ (%)	82.59	8.96	5.19	1.35	0.58

^a PS: potassium sorbate, BHA: butylated hydroxyanisole, Glu: glucose

3.2. Back-propagation neural network

Inspired by the neural network behaviors of learning and information processing in the human brain, the back-propagation neural network constitutes a nonlinear dynamical framework to train multi-layer feedback in a self-learning way. In a typical neural network, there is an input layer, an output layer and hidden layers among them [24]. For the input vector $x(x_1, x_2, \dots, x_n)$, the j th neuron of the hidden layer can be expressed as [34],

$$b_j = f_1 \left(\sum_{i=1}^n \omega_{ij} x_i - \theta_j \right) \quad (2)$$

where f_1 is the transfer function of the hidden layer, which is usually in the form of the S-shaped sigmoid curve [24]. θ_j is the threshold value of the j th neuron in hidden layer, ω_{ij} represents the weight from the input

layer to the hidden layer. Similarly, the output of the network y_k can be calculated by [34],

$$y_k = f_2 \left(\sum_{j=1}^s \omega_{jk} b_j - \theta_k \right) \quad (3)$$

where f_2 is the transfer function of the output layer, θ_k is the threshold value of the output layer, the weight from the hidden layer to the output layer is ω_{jk} .

Supposing the expected output is t_k and the actual output from the network is y_k , the mean squared error function is defined as [34],

$$\varepsilon = \frac{1}{m} \sum_{k=1}^m (t_k - y_k)^2 \quad (4)$$

The network training is then self-learning by constantly adjusting and modifying weights and thresholds (error). Particularly, the weight is inversely modified from the output layer to the hidden layer and then from the hidden layer to the input layer until achieving the minimum error or maximum training step. Finally, the samples to be predicted are input into the trained network, and then the predicted results are obtained. In this work, the number of hidden layer neurons was set to 3, the mean squared error was set as 0.001, and the Levenberg–Marquardt algorithm [35] was used for training within 1000 iterations.

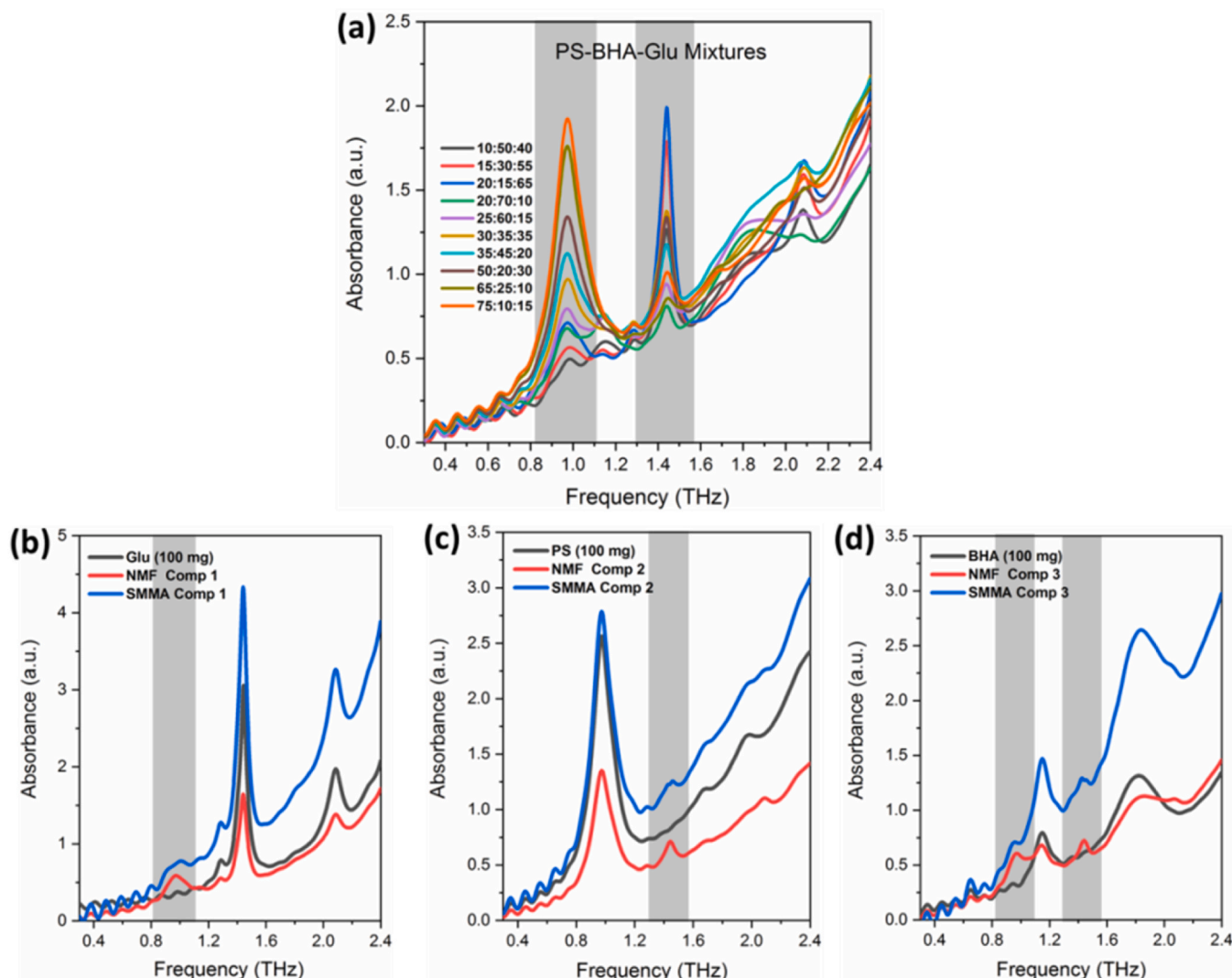


Fig. 3. (a) THz spectra of ternary mixtures, (b) – (d) comparison of the extracted component (Comp) spectra and measured spectra of pure reagents.

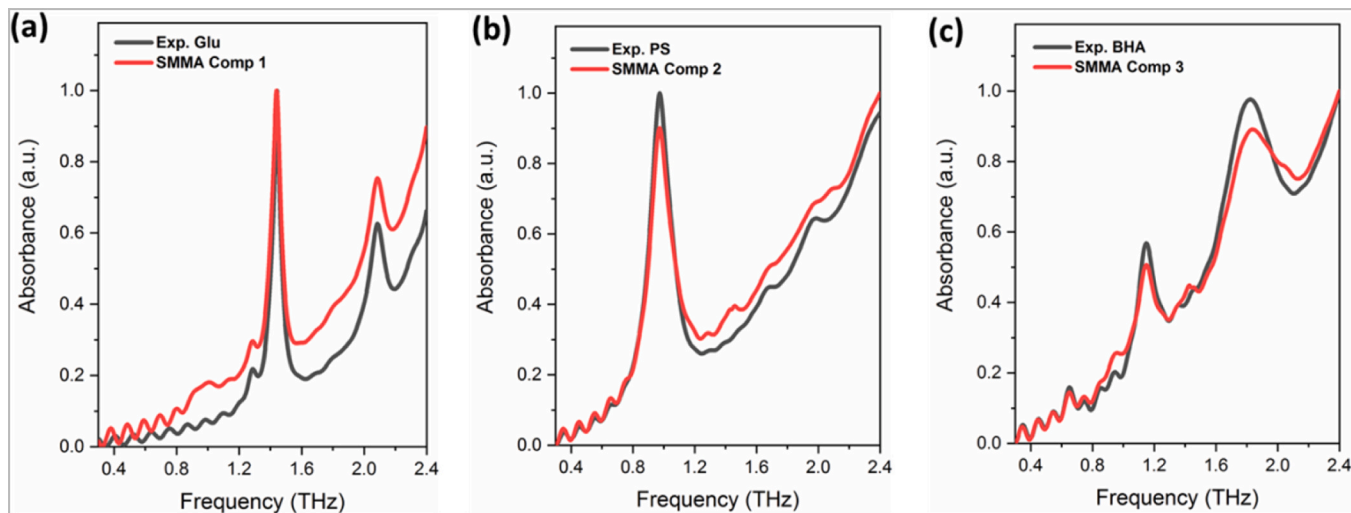


Fig. 4. Comparison of standardized SMMA component spectra and standardized experimental (Exp.) spectra.

4. Results and discussion

4.1. Number of Components

Acquiring component number k from unknown mixtures is a crucial prerequisite for further analysis. Because the ordered singular values (SVs) of the original spectral matrix $V_{m \times n}$ provide a quantitative estimation of the representation in the data matrix [36], they are applied to determine the number of component spectra (k). Normally, the first few significant SVs are supposed to correlate with usable components, while the subsequent smaller SVs are ascribed to the contributions of noise [37,38]. To quantitatively estimate the singular value, the relative contribution rate (RCR) is defined as:

$$\gamma_l = \frac{\sigma_l}{\sum_{l=1}^n \sigma_l} \quad (5)$$

where σ_l is a singular value for $l = 1, 2, \dots, n$. Empirically, 2.5% is set as the RCR threshold [39,40], and the number of SVs with RCR greater than 2.5% is evaluated as the component number k of the mixed system, while SVs with RCR less than 2.5% have been ignored as system noise.

The SVD of spectral matrix $V_{m \times n}$ is listed in Table 3, which shows that singular values decrease drastically, and their corresponding RCRs are listed below. The RCRs of the first three significant SVs are more than 2.5%, which are supposed to interpret usable components. Thus, SVD determines the mixtures containing three components ($k = 3$), which

well matched with the actual ternary mixtures. Moreover, in binary mixture analysis, SVD also accurately obtains the component number [13].

4.2. Component spectra extraction

THz spectra of mixtures (PS, BHA and Glu) comprised of ten mixed ratios are shown in Fig. 3(a). Obviously, two prominent absorption features at 0.97 THz and 1.44 THz are displayed in all mixed samples (gray area), which originate from PS and Glu, respectively. Specifically, the absorption feature at 0.97 THz is mainly derived from intramolecular vibrations of PS molecules [41], while the absorption feature at 1.44 THz comes from intermolecular vibrations of Glu molecules, especially hydrogen-bonded interactions [42,43]. Moreover, the significant absorption features exhibit linear intensity with the content of pure reagent [44]. However, other absorption features overlap and become blurry. Particularly, the weak and broad THz features of BHA were indistinguishable in the mixtures' spectra. Then, SMMA and NMF are utilized to decompose the spectral matrix of the mixtures. As shown in Fig. 3(b) – (d), three individual component spectra are successfully extracted and compared with the measured spectra of pure reagents.

However, the absorption features extracted by NMF are much weaker than the experimental ones, especially for the significant peaks such as peaks at 0.97 and 1.44 THz. Moreover, NMF based on mathematical decomposition seems to inadequately resolve the componential features at the positions of the prominent peaks in mixtures, and a weak

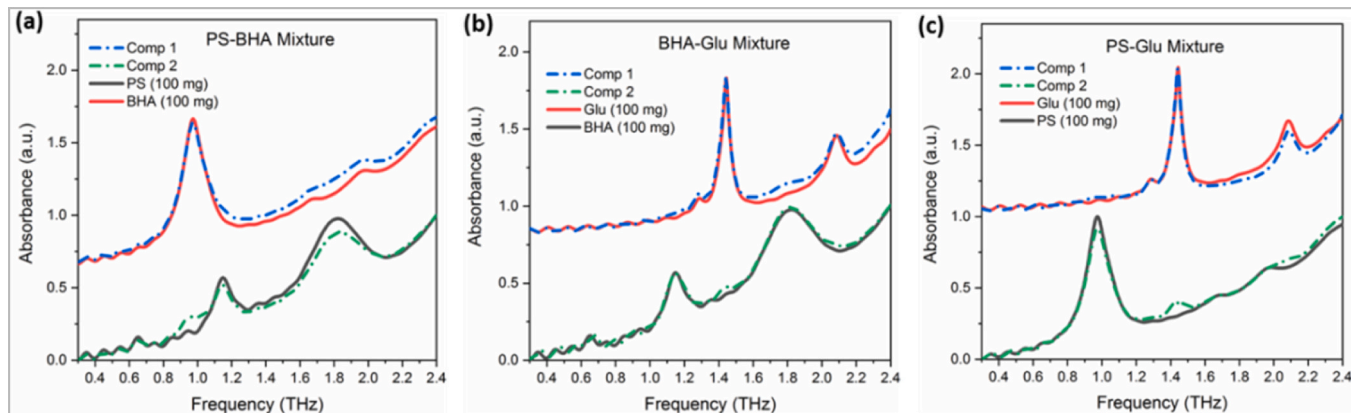


Fig. 5. The extracted component spectra (standardized) from (a) PS-BHA (b) BHA-Glu and (c) PS-Glu mixtures.

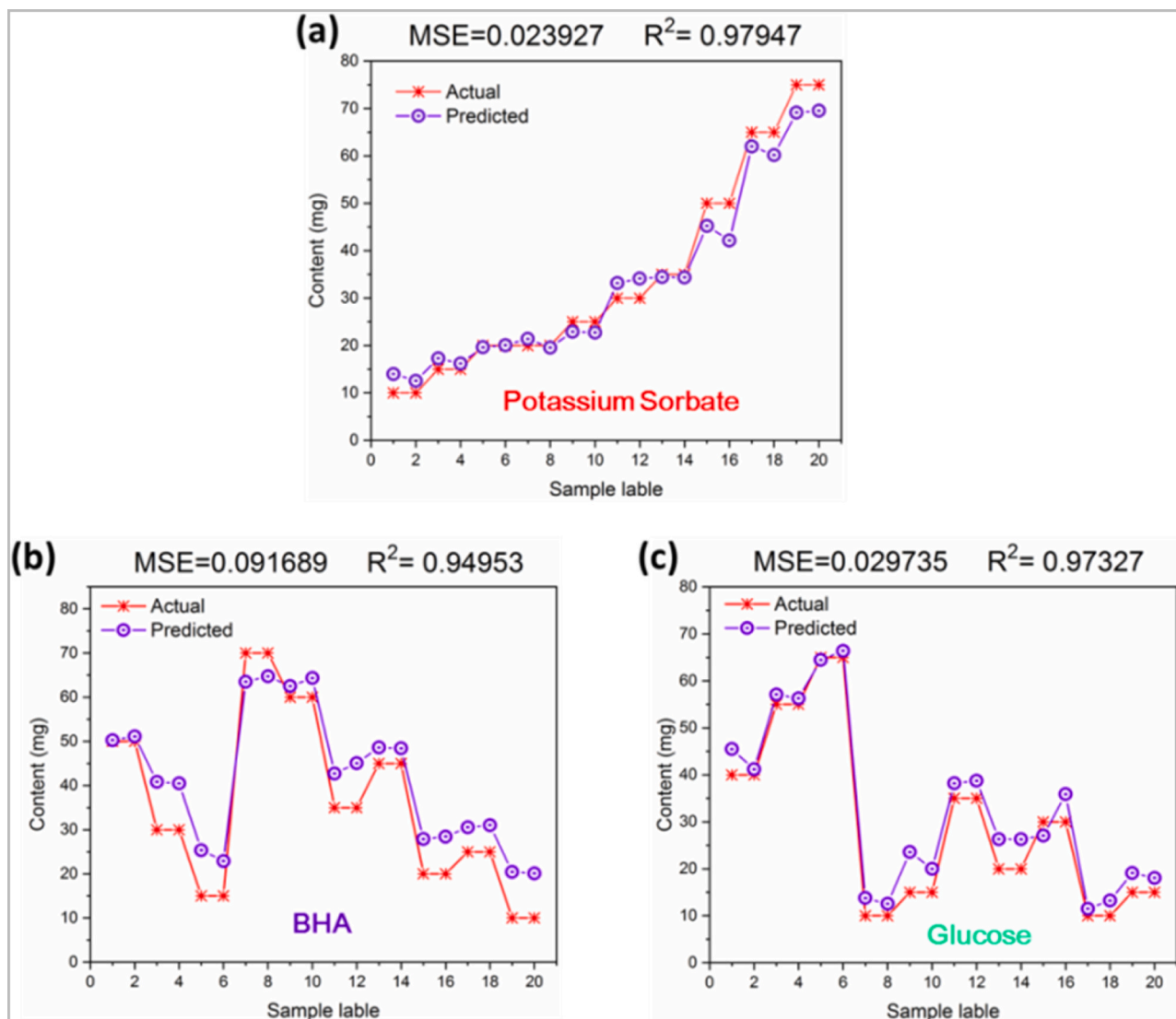


Fig. 6. SVR predictions against actual contents.

absorption peak is additionally stripped out (see also gray area in Fig. 3 (b) – (d)). It is more serious for BHA in Fig. 3(d), with two comparable additional peaks (gray area) extracted by NMF, which causes prominent discrepancy between the NMF extracted spectrum and experimental one.

For the spectra extracted by SMMA, the intensities of absorption features are identical as the experimental ones. Therefore, compared to the spectra extracted by NMF (much weaker), the spectra extracted by SMMA are more consistent with the experimental spectra of pure reagents. However, another serious issue is that the baselines of the spectra extracted by SMMA drift obviously, especially for BHA. To solve baseline drift, the measured spectral vector x and extracted spectral vector $\hat{x}$ were preliminarily standardized as x' and $\hat{x}'$, respectively [45],

$$x'_i = \frac{(y_{\max} - y_{\min}) \cdot (x_i - x_{\min})}{(x_{\max} - x_{\min})} + y_{\min} \quad (6)$$

$$\hat{x}'_i = \frac{(y_{\max} - y_{\min}) \cdot (\hat{x}_i - \hat{x}_{\min})}{(\hat{x}_{\max} - \hat{x}_{\min})} + y_{\min} \quad (7)$$

where i represents 701 variables for each spectrum. Therefore, the measured spectra and the spectra extracted by SMMA are normalized to the same absorbance interval $[y_{\min}, y_{\max}]$ set as $[0, 1]$. As shown in Fig. 4, after standardization, the baselines of the spectra extracted by SMMA

drift consistently with the experimental spectra of pure reagents. In particular, compared to Fig. 3(d), the baseline correction is more significant for BHA, as shown in Fig. 4(c).

Moreover, to further verify the baseline correction by standardization, binary mixtures containing any two of the food additives (PS, BHA and Glu) were prepared. PS and BHA, BHA and Glu, BHA and Glu mixtures, 27 in total mixed samples with different mass ratios were measured, and their spectral matrices were decomposed by SMMA. As shown in Fig. 5, after standardization, the extracted spectra almost coincide with the measured spectra, further indicating that SMMA effectively extracted component spectra from multicomponent mixtures. Ultimately, once the component spectra were extracted from complicated mixtures and compared to distinctive vibrational features of substances in the THz region, it achieved accurate and rapid identification of the compositions in mixtures.

4.3. Quantitative analysis

The intensity of the characteristic absorption peak is sensitive to the content of the pure reagent according to the Beer–Lambert law [46]. However, for complicated multicomponent mixtures, it is almost impossible to directly obtain the component content from seriously overlapping spectra. The relationship between the mixed spectral

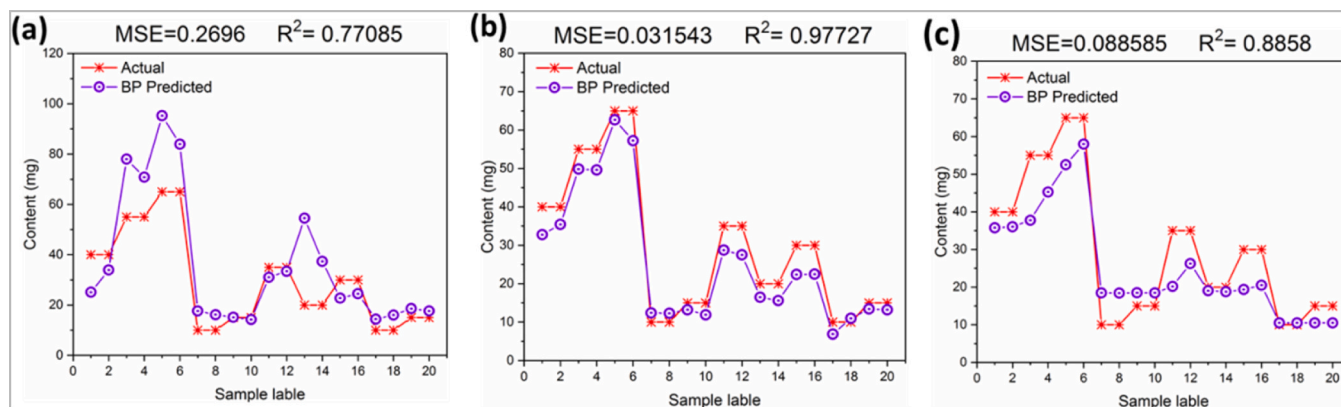


Fig. 7. BPNN predictions in three random executions.

features and component content can only be mined by machine learning. SVR is a supervised learning algorithm used to find a regression model from features to a target by mapping to a higher dimensional linear space. And SVR achieves accurate predictions in complex small- and medium-scale nonlinear problems [21]. In this work, the SVR model using a radial basis function kernel was implemented in the LIBSVM library [47]. The regression model from the spectral features to target component content was trained by 80 experimentally measured spectra and validated by 20 testing datasets (labeled from 1 to 20).

For a visualized comparison, SVR predictions against the actual contents are shown in Fig. 6, where the predicted contents agree well with the actual contents with slight deviations. Moreover, the mean square error (MSE) and decision coefficient (R^2) were used to evaluate the performance of SVR predictions. The low MSE and high R^2 indicate more reliable and accurate predictions. It shows SVR predictions generate MSE = 0.0239, 0.0917, 0.0297 and R^2 = 0.9795, 0.9495, 0.9733 for PS, BHA and Glu, respectively. Moreover, it is worth noting that original spectral data without baseline calibration are used for the SVR model, but the predictions still achieve satisfied results, indicating the robustness and accuracy of the SVR algorithm.

On the other hand, BPNN was also applied to predict the component contents. However, as shown in Fig. 7, the results predicted by BPNN are quite different in every execution. Specifically, the BPNN predictions can achieve the comparable decision coefficient of R^2 = 0.9727 as SVR, or lower decision coefficients of R^2 = 0.7085 and R^2 = 0.8858 in the other two random executions. The reason may be that a relatively stable and accurate neural network cannot be obtained with a small training dataset by the high nonlinearity BPNN. Conventionally, the input data are randomly divided into a training set and a testing set in BPNN. Approximately two-thirds of the data are applied to train the network [48]. And the weights of the neurons are assigned randomly in initialization. When the training pattern is imported to the neural network, the network is learned by modifying and adjusting the weights. Therefore, to obtain stable and accurate predictions, the training dataset must provide as many of the representative samples which contain a variety of distinct characteristics from the problems that may be encountered in the final application of the neural network. The risk of undersampling for nonlinear functions can be significantly decreased by a large training set and BPNN can achieve good performance for complex systems [48].

5. Conclusions

Compared with principal component analysis, which locates the samples to cluster in the score plots for identification purposes [10,49], the method of extracting component THz spectra from the mixtures is intuitive and effective [11,40]. In this work, the difficulties of the NMF and SMMA encountered in extracting component spectra from more complicated ternary mixtures were solved. NMF was initialized by a

pure variable approach to obtain the unique decomposition results. Compared to NMF [50], SMMA is more suitable to extract component spectra from multicomponent mixtures. After standardization to correct baseline drift, the spectra extracted by SMMA are highly consistent with the measured spectra, which greatly facilitate the rapid identification of compositions in mixtures. On the other hand, SVR predicted the contents of each individual component with high robustness and the decision coefficient R^2 greater than 0.949. Therefore, compared with the traditional PLS and improved PLS methods [51,52], SVR exhibits accurate, robust and reliable predictions in mixed quantitative analysis [52,53]. Additionally, machine learning methods were also applied to improve the SNR of THz spectra and then improve the quantitative accuracy [51]. However, no machine learning algorithm is of universal applicability, and they must be improved continuously for specific problems. It is believed that machine learning algorithms play an increasingly important role in complex mixture analysis, and fingerprint THz spectroscopy combined with machine learning shed light on the urgent demand in actual scenario detection of mixtures.

CRedit authorship contribution statement

Hui Yan: Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Investigation. **Wenhui Fan:** Conceptualization, Methodology, Supervision, Funding acquisition. **Chong Qin:** Data curation, Visualization. **Xiaoqiang Jiang:** Formal analysis, Validation. **Yuming Zhang:** Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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